LLM Evaluation for Intelligent Document Workflows

NLP for Business and Finance Project

Giacomo Lugana, Roberto Stoian

May 2026

Abstract

This report presents a comparative evaluation of open-source vision-language models (VLMs) for two core invoice-processing tasks: document classification and structured-field extraction. We benchmark zero-shot and few-shot strategies across six models on the RVL-CDIP dataset (16-class document classification) and the SROIE dataset (receipt field extraction). We further fine-tune DeiT-small and PaliGemma2-3B with LoRA on the classification task. Results show that specialised fine-tuned models dominate zero/few-shot VLMs on classification, while Qwen2-VL-7B achieves the strongest extraction performance among off-the-shelf models. Fine-tuned PaliGemma2-3B reaches 90.0% accuracy and 0.882 invoice F1, approaching the specialist DiT-large at a fraction of the parameters. Few-shot prompting consistently hurts classification but benefits large-model extraction, underscoring that the optimal inference strategy is task- and model-dependent.

1 Introduction

The steady and fast advancements in the field of Artificial Intelligence are there for everyone to see. The full potential of this technology has not been reached yet, nevertheless it is already reshaping the way people work. In the business context, intelligent document processing is a key enabler for enterprise automation. The project, in partnership with Sacco AI Consulting, focuses on document understanding using NLP and LLMs. The works attempts to implement a rigorous framework of what open-source, locally deployable models can currently achieve.

The rapid progress of open-source multimodal language models – Qwen2-VL, PaliGemma, SmolVLM, and others – has opened the possibility of building capable document pipelines without cloud dependencies. However, most public benchmarks evaluate these models on natural images or general-purpose tasks. Business documents present a different challenge: they are predominantly text-heavy, structurally repetitive within a class but visually similar across classes, and they might be scanned, hand-written, but also digital-born. It is therefore not obvious whether strong performance on natural image benchmarks translates to enterprise document workflows. Moreover, a great amount of business documents contain sensitive data, excluding apriori very powerful yet "black-box" models; this choice can be justified (at least) by two reasons: safety and transparency.

We addressed two operational tasks that mirror a realistic processing pipeline: (1) routing incoming documents to the correct downstream destination via *document classification*, and (2) populating structured databases via *key-field extraction*. For each task we tested both zero-shot and few-shot prompting strategies, and we fine-tuned two architecturally distinct models to quantify the gap that targeted training can close relative to off-the-shelf inference.

2 Main Contributions - DA RIVEDERE ANCORA

1. **Dataset exploration.** Characterised the RVL-CDIP image corpus (16 classes, 500 samples per class) and identified visual properties — brightness, aspect ratio, layout structure — relevant to model selection.

2. **Document classification.** Evaluated six models under zero-shot and 1-shot (32 labelled examples) strategies on a 1,600-sample test set, reporting accuracy, macro F1, and invoice-specific F1, with DiT-large as a near-optimal upper-bound reference.

3. **Information extraction.** Applied zero-shot and 10-shot prompting to extract four key fields from SROIE receipts, benchmarking against a Tesseract OCR + regex baseline across three metrics.

4. **Fine-tuning.** Trained DeiT-small (two-phase head + partial backbone) and PaliGemma2-3B with LoRA on 32,000 images, measuring the gain over zero-shot VLM baselines and quantifying the efficiency of parameter-efficient adaptation.

3 Materials and Methods

The experiments are built on two publicly available benchmarks, each targeting one of the two tasks. Both are widely adopted in the document understanding literature and serve as standard reference points for evaluating model performance.

3.1 Datasets

RVL-CDIP (Classification task). The Ryerson Vision Lab Complex Document Information Processing dataset [1] contains roughly 400,000 grayscale document images across 16 classes, spanning a wide range of business and administrative document types: *advertisement*, *budget*, *email*, *file folder*, *form*, *handwritten*, *invoice*, *letter*, *memo*, *news article*, *presentation*, *questionnaire*, *resume*, *scientific publication*, *scientific report*, and *specification*.
We streamed a balanced subset of 8,000 images (500/class) for exploration and model selection, and drew 32,000 images (2,000/class) for fine-tuning. Evaluation used a standardised test set of 1,600 images (100/class). All images come in grayscale TIFF format, converted to JPEG/PNG for model inference.

SROIE (Extraction task). The Scanned Receipts OCR and Information Extraction dataset [2] contains approximately 970 coloured receipt images in JPEG format. All images originate from Malaysian retail environments. Ground truth is provided at the word level via NER-style annotations; we reassemble contiguous tagged words into complete strings to obtain 4 field values per receipt:

- `company`: the store or vendor name;
- `date`: transaction date;
- `address`: store address;
- `total`: final amount due.

Before computing any metric, both model predictions and ground truth are lowercased, punctuation-normalised, and whitespace-collapsed to reduce sensitivity to formatting differences. In terms of size, the test split features 347 items.

3.2 Models

We selected models to cover a range of architectures and parameter budgets, from 500M to 7B, enabling a comprehensive cost-performance analysis. The four VLMs (Qwen2-VL-7B, Qwen2.5-VL-3B, PaliGemma2-3B, SmolVLM-500M) are general-purpose instruction-tuned models capable of reading and reasoning about image content in response to free-text prompts.
Table 1 lists all models evaluated. As a side note, DiT-large is a document-specific vision transformer that was trained and fine-tuned directly on RVL-CDIP; it serves as a near-optimal specialist reference rather than a fair competitor to the other models. CLIP ViT-B/32 encodes both the image and text class labels into a shared embedding space and classifies by cosine similarity – a simple but efficient baseline.

Table 1: Models evaluated across tasks.

Model	**Params**	**Type**	**Task**
DiT-large [3]	307M	Doc. ViT (specialist)	Classification (upper bound)
DeiT-small [4]	22M	ViT	Classification (fine-tuned)
PaliGemma2-3B [5]	3B	VLM	Classification, Extraction
Qwen2-VL-7B [6]	7B	VLM	Classification, Extraction
Qwen2.5-VL-3B [6]	3B	VLM	Classification, Extraction
SmolVLM-500M [8]	500M	VLM	Classification, Extraction
CLIP ViT-B/32 [7]	151M	CLIP	Classification

3.3 Experimental Setup

3.3.1 TASK #1 - CLASSIFICATION

Classification – zero-shot vs few-shot. Each VLM received a structured prompt presenting the 16 class names as a numbered list and asking to return exactly one label. To minimise prompt sensitivity, we fixed a single prompt template. The prompt text was kept as consistent as possible across models, while accounting for their respective structural differences. DiT-large produces 16-dimensional class logits directly from the image encoder, without any natural-language prompt. All models were evaluated on the same 1,600-image test set with a fixed random seed.
For the few-shot setting, we selected 2 representative images per class from the training split (32 images total) and attached them as in-context examples with their labels before the query image. The same 32 images were used for all models to ensure comparability.

Fine-tuning – DeiT-small, PaliGemma2-3B with LoRA [9]. DeiT-small [4] is a vision transformer with 22M parameters, pretrained on ImageNet using knowledge distillation. Its pretraining is completely general-purpose, with no prior knowledge of documents. We fine-tune it in two phases: in Phase 1 (3 epochs, LR $= 3 \times 10^{-4}$, cosine schedule) the backbone is frozen and only a new classification head is trained; in Phase 2 (5 epochs, LR $= 5 \times 10^{-5}$) the top 4 transformer blocks are unfrozen to let the model adapt its higher-level features while keeping the lower layers unchanged. The training set consisted of 32,000 balanced images (2,000/class).
PaliGemma2-3B is a vision-language model with 3B parameters. Since fine-tuning all parameters would be too expensive on available hardware, we use Low-Rank Adaptation (LoRA) adapters (r=16, α=32, dropout = 0.05), which are small modules inserted into the attention layers of the language model backbone. This keeps the number of trainable parameters small relative to the full model size. Phase 1 (3 epochs, LR $= 3 \times 10^{-4}$) trains only the classification head with the backbone frozen; Phase 2 (5 epochs, LR $= 5 \times 10^{-5}$) also enables the LoRA adapters.

3.3.2 TASK #2 - EXTRACTION

Extraction – zero-shot vs few-shot. The extraction task workflow mirrors the same structure as the preceding classification procedure. The baseline uses Tesseract OCR to extract raw text from each receipt image, followed by hand-crafted regular expressions for each of the four fields. For VLMs, we designed a structured prompt asking the model to return a JSON object with keys `company`, `date`, `address`, and `total`. Model outputs were then parsed with a robust regex to handle malformed or partial JSON.
For few-shot, 10 fully annotated receipt examples were prepended to the prompt before the query image. Due to its relatively limited context window, only 4 examples were given in the prompt to SmolVLM-500M. The results were then compared with the ground truth to compute the metrics.

3.3.3 EVALUATION METRICS

Classification and extraction metrics. Once the results were obtained, defining appropriate evaluation metrics proved essential to reliably assess model performances and draw meaningful insights.
For classification, we report: overall accuracy, macro-averaged F1 across all 16 classes, and the binary F1 specifically for the *invoice* class.

$$\text{Accuracy} = \frac{\text{Correct predictions}}{1600} \tag{1}$$

$$F1_{\text{macro}} = \frac{1}{16}\sum_{i=1}^{16} F1_i, \qquad \text{where } F1_i = 2 \cdot \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{2}$$

$$F1_{\text{invoice}} = 2 \cdot \frac{\text{Precision}_{\text{invoice}} \times \text{Recall}_{\text{invoice}}}{\text{Precision}_{\text{invoice}} + \text{Recall}_{\text{invoice}}} \tag{3}$$

For extraction, we follow and expand the SQuAD evaluation protocol [**?**], a widely adopted benchmark originally designed for reading comprehension tasks, which evaluates predictions against ground truth answers at the token and character level.
Specifically, we provide: Exact Match (EM=1 if prediction equals ground truth exactly), Token F1 (TF1, word-level precision/recall harmonic mean), and Character F1 (CF1, character-level). Such range of metrics allows to move from strict to progressively more lenient matching criteria.

$$EM = \mathbf{1}[\hat{y} = y] \tag{4}$$

$$TF1 = 2 \cdot \frac{P_{\text{token}} \times R_{\text{token}}}{P_{\text{token}} + R_{\text{token}}}, \qquad \text{where } P_{\text{token}} = \frac{|\hat{T} \cap T|}{|\hat{T}|}, \quad R_{\text{token}} = \frac{|\hat{T} \cap T|}{|T|} \tag{5}$$

$$CF1 = 2 \cdot \frac{P_{\text{char}} \times R_{\text{char}}}{P_{\text{char}} + R_{\text{char}}}, \qquad \text{where } P_{\text{char}} = \frac{|\hat{C} \cap C|}{|\hat{C}|}, \quad R_{\text{char}} = \frac{|\hat{C} \cap C|}{|C|} \tag{6}$$

where $\hat{y}$ is the predicted answer and y is the correct answer; $\hat{T}$ and T are the sets of words in the predicted and correct answer; $\hat{C}$ and C are the sets of characters in the predicted and correct answer.

4 Results

4.1 Preliminary EDA

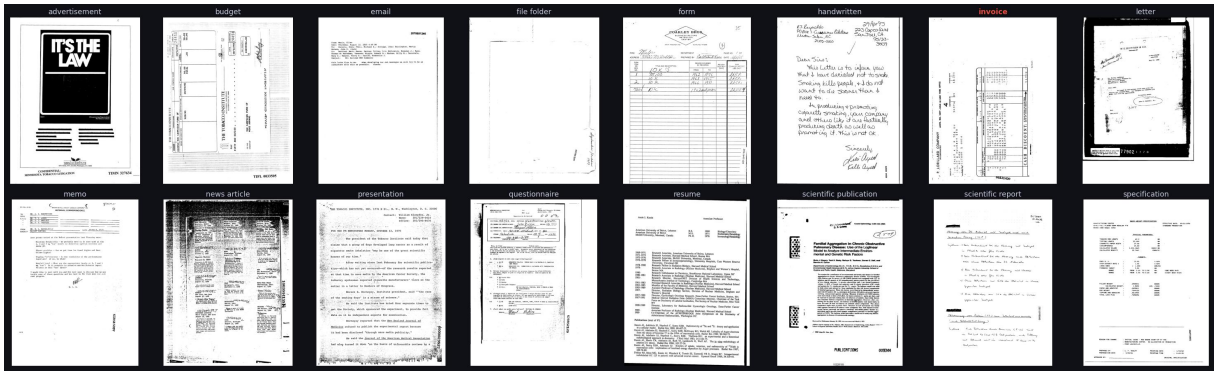


Figure 1: One sample per class from RVL-CDIP

The balanced subset contains 500 items per class, for a total of 8,000 images. All images share a near-identical aspect ratio (766 × 1000 px, ratio = 0.77) and are in grayscale. Invoice images have a mean brightness of 237.8/255, consistent with white-background scanned documents. Figure 1 reveals several important observations for model selection.

First, a number of class pairs share highly similar visual layouts: *form* and *questionnaire* both feature structured grids of text fields; *memo* and *letter* share single-column dense text; and *scientific publication* and *scientific report* both use two-column layouts with references. These similarities create hard disambiguation cases that require fine-grained text understanding rather than coarse visual pattern matching.

Second, the invoice class is visually distinctive – featuring row-column numerical tables, totals at the bottom, and header information – which explains why several VLMs achieve higher invoice F1 score than their overall macro F1. The feature heatmap (Appendix A.4) confirms that invoice images cluster tightly in pixel-space, supporting the effectiveness of targeted fine-tuning.

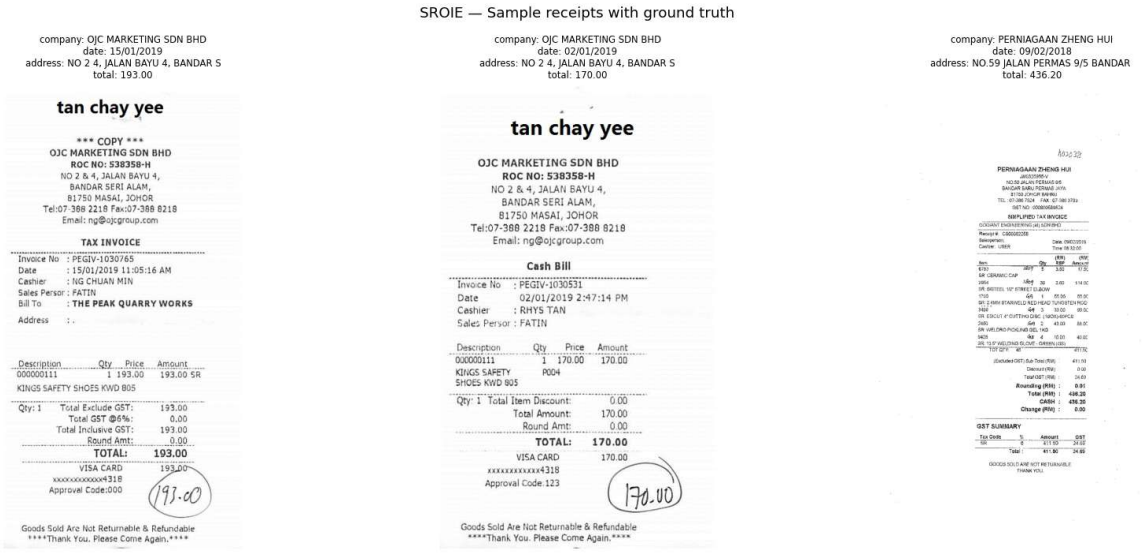


Figure 2: Sample receipts from SROIE with annotations

The SROIE corpus features 973 colour JPEG receipts (626 training, 347 test). Differently from RVL-CDIP, images are RGB and exhibit substantial size variability: typical dimensions

are approximately 943 × 1616 px (aspect ratio = 0.58, portrait-oriented), but receipt heights range from 605 to over 2,300 px, reflecting very different paper lengths and scan setups. Mean brightness (219.3/255) is lower than for RVL-CDIP invoice images, consistent with denser ink coverage and occasional coloured logos or backgrounds. Field-level difficulty is apparent even from the OCR baseline results. `date` is the easiest field (regex exact match: 0.70), since its format is highly regular (DD/MM/YYYY or similar short patterns that hand-crafted rules can capture reliably). `company` and `address` fields are the hardest: company names combine capitalisation variants, abbreviations, and OCR noise, making the regex baseline nearly ineffective (EM = 0.112), while addresses span multiple lines with inconsistent abbreviations and postal codes (EM = 0.063). `total` falls in between (EM = 0.288), hindered mainly by optional currency symbols and decimal separators. This field-level analysis motivates the use of character-level F1 (CF1) as an interesting additional metric, since partial token matches carry meaningful signal for multi-word fields such as address – a field where even strong VLMs rarely achieve exact match.

4.2 Document Classification

4.2.1 Zero-Shot Results

Model	Accuracy	Macro F1	Invoice F1
DiT-large (specialist, upper bound)	0.929	0.929	0.935
Qwen2-VL-7B	0.546	0.514	0.693
Qwen2.5-VL-3B	0.541	0.501	0.604
PaliGemma2-3B	0.418	0.369	0.241
CLIP ViT-B/32	0.287	0.263	0.123
SmolVLM-500M	0.266	0.227	0.381

Table 2: Zero-shot classification on RVL-CDIP (1,600 test images)

DiT-large reaches 92.9% accuracy, confirming that a specialist model pretrained and fine-tuned on the target distribution sets a hard-to-match upper bound. Among general VLMs, Qwen2-VL-7B and Qwen2.5-VL-3B achieve ≈ 54% accuracy – a large gap relative to DiT-large, but substantially above chance (6.25%, which equals 1/16). PaliGemma2-3B reaches only 41.8%, largely because it assigns high confidence to a small subset of classes, yielding near-zero F1 on several categories (see confusion matrix in Appendix A.3). CLIP and SmolVLM perform at or below 29%; CLIP suffers because its text-image similarity struggles to discriminate classes whose visual appearance is not aligned with simple label names, while SmolVLM's 500M capacity appears insufficient for reliable instruction following on a 16-way classification task. Notably, invoice F1 for Qwen2-VL-7B (0.693) is 18 points above its macro F1 (0.514), confirming that the invoice class has distinctive visual features that VLMs can leverage even without task-specific training.

4.2.2 Few-Shot Results

Model	Accuracy	Macro F1	Invoice F1
DiT-large (specialist, upper bound)	0.930	0.930	0.925
Qwen2.5-VL-3B	0.501	0.493	0.506
Qwen2-VL-7B	0.389	0.378	0.612
CLIP ViT-B/32	0.354	0.313	0.365
PaliGemma2-3B	0.063	0.007	0.000

Table 3: Few-shot document classification results

Adding 32 in-context examples produced mixed and predominantly negative results for classification. For most models, zero-shot outperforms few-shot. This collapse in prediction diversity is a known limitation of smaller VLMs: when the input context is too long, the model fails to attend to the relevant visual features and instead relies on superficial patterns present in the examples. The clearest failure is PaliGemma2-3B, which collapses to predicting *scientific report* for virtually every input (accuracy 0.063, macro F1 0.007). Such outcome was predictable as PaliGemma2-3B does not natively support multi-turn conversation or multi-image interleaved prompting, making it structurally incompatible with the few-shot setting. Qwen2-VL-7B accuracy drops from 0.546 to 0.389; the 7B model is more robust but still shows attention dilution when the context is heavily padded with images. Qwen2.5-VL-3B is relatively stable (-0.04 accuracy), suggesting that its shorter-context architecture is better suited to this setting. CLIP is the only model that improves in few-shot: the additional text descriptions provide stronger semantic anchoring for its contrastive matching. The delta analysis (Appendix A.3) highlights that for classification, the zero-shot technique is strictly preferred for most chosen models.

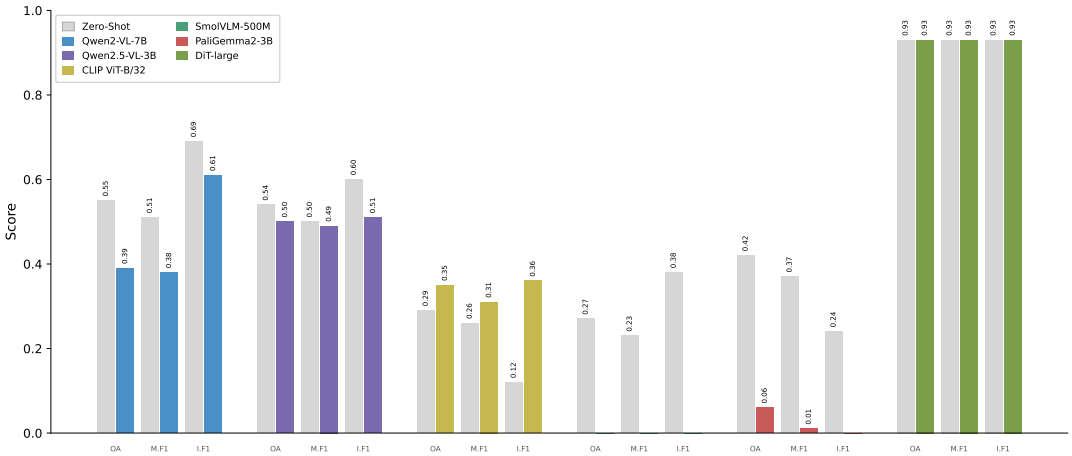


Figure 3:
Zero-shot (grey) vs. few-shot (coloured) classification across all models.
OA = Overall Accuracy, M.F1 = Macro F1, I.F1 = Invoice F1

4.3 Information Extraction

	Zero-shot			10-shot		
Model	EM	TF1	CF1	EM	TF1	CF1
Qwen2-VL-7B	0.477	0.660	0.820	**0.501**	**0.696**	**0.888**
PaliGemma2-3B	0.433	0.607	0.795	0.077	0.150	0.642
Qwen2.5-VL-3B	0.345	0.498	0.656	0.311	0.481	0.750
SmolVLM-500M	0.308	0.467	0.609	0.000	0.000	0.000
Baseline (OCR+regex)	0.291	0.441	0.632	—	—	—

Table 4: Extraction results on SROIE — overall metrics (average across all four fields). EM = Exact Match, TF1 = Token F1, CF1 = Character F1.

The extraction results reveal a clear size-dependent pattern with respect to few-shot prompting. Qwen2-VL-7B is the best extractor in both regimes and improves substantially with 10-shot context (overall CF1: 0.820 → 0.888). Qwen2.5-VL-3B also improves (0.656 → 0.750), though from a lower baseline. Conversely, PaliGemma2-3B degrades significantly (0.795 → 0.642 overall CF1): the same prompt-sensitivity that caused mode collapse in classification is here manifested as a formatting failure – the model stops producing well-formed JSON when examples are present, breaking the output parser. SmolVLM-500M fails entirely in 10-shot mode (CF1 = 0.000 across all fields), a complete breakdown attributable to context length saturation.

At the field level, `address` is consistently the hardest field for all models: addresses span multiple lines, use non-standard abbreviations, and their format varies widely across the Malaysian receipts in SROIE. In contrast, `date` benefits from highly regular patterns (DD/MM/YYYY or similar), which is why both the regex baseline and VLMs perform well on it. `company` name extraction is where VLMs most clearly outperform the regex baseline: exact company names are nearly impossible to match with hand-crafted rules when capitalization, abbreviations, and OCR noise interact, while VLMs can leverage visual context and language understanding.

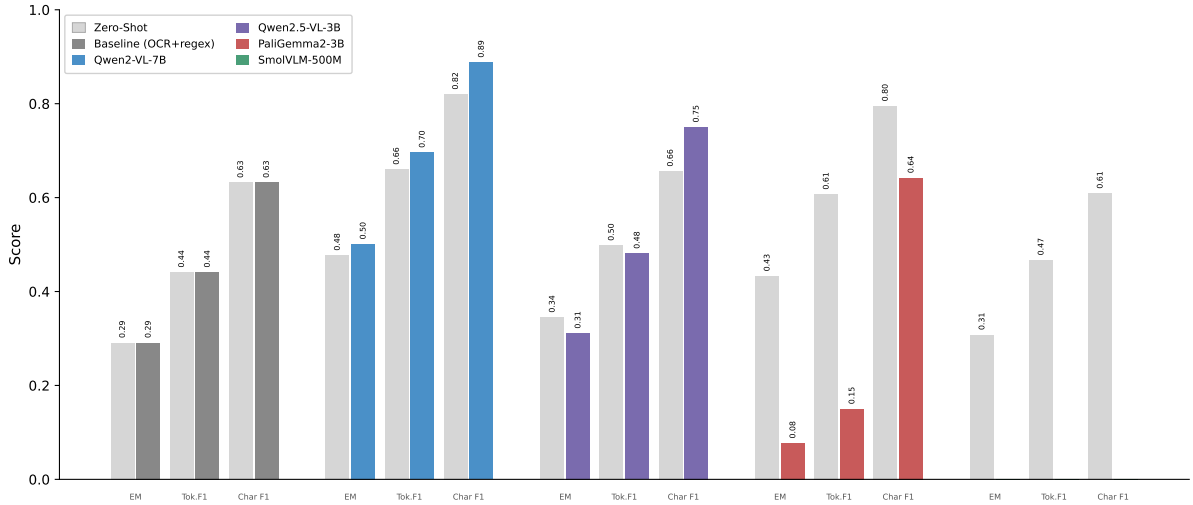


Figure 4:
Extraction results (EM, Token F1, Char F1) across models

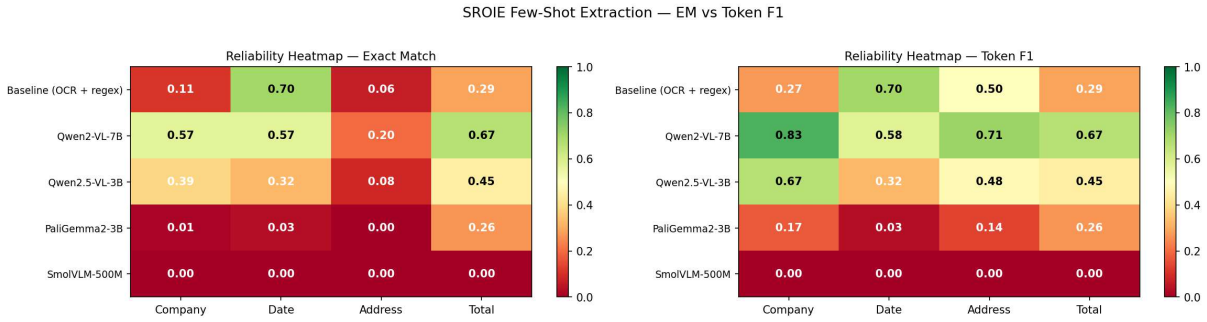


Figure 5: Extraction reliability heatmap and zero/few-shot comparison

4.4 Fine-Tuning for Classification

Model	Type	Acc.	Macro F1	Invoice F1
DiT-large	specialist	0.930	0.930	0.925
PaliGemma2-3B (LoRA FT)	fine-tuned	0.900	0.885	0.882
DeiT-small (FT)	fine-tuned	0.757	0.741	0.600
Qwen2-VL-7B	zero-shot	0.546	0.514	0.693
Qwen2.5-VL-3B	zero-shot	0.541	0.501	0.604
PaliGemma2-3B	zero-shot	0.418	0.369	0.241
DeiT-small	pre-trained (no FT)	0.060	0.036	0.000

Table 5: Fine-tuned models vs. zero-shot baselines on RVL-CDIP

Fine-tuning produced by far the largest performance gains. DeiT-small goes from 6.0% accuracy (random-level performance – its ImageNet pretraining provides no document understanding) to 75.7% after fine-tuning, a gain of +69.7 percentage points overall and +0.643 invoice F1. This is particularly striking given the model's size: 22M parameters trained in two phases on a single GPU outperforms every zero-shot and few-shot VLM tested, including models up to 14× larger. The two-phase training strategy is central to this result: training only the head in Phase 1 provides a stable initialisation before opening the backbone in Phase 2, which prevents the detrimental forgetting that would result from immediately fine-tuning all layers at a high learning rate.

PaliGemma2-3B with LoRA reaches 90.0% accuracy and 0.882 invoice F1, closing to within 3 percentage points of DiT-large – a model specifically pretrained on the target dataset. This is achieved while training only a small fraction of the total parameters (the LoRA adapters and the classification head), demonstrating that parameter-efficient fine-tuning is a practical and cost-effective adaptation strategy. On average, PaliGemma2-3B gains +55 percentage points across metrics from its zero-shot baseline, and DeiT-small gains +67 percentage points.

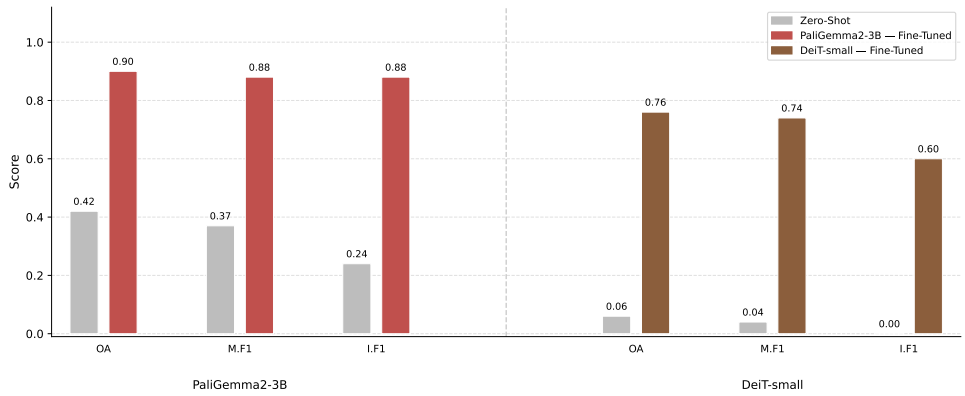
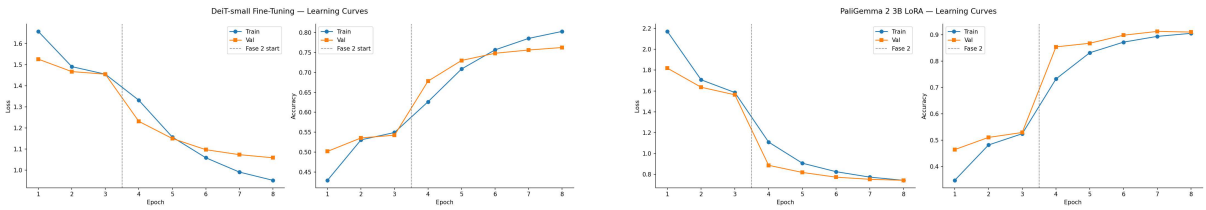


Figure 6: Fine-tuned vs. zero-shot performance for PaliGemma2-3B and DeiT-small



(a) DeiT-small training curves.

(b) PaliGemma2 LoRA training curves.

Figure 7: Training and validation loss/accuracy during fine-tuning

5 Conclusions

This project evaluated whether open-source, locally deployable models can reliably support an enterprise intelligent document platform for invoice processing. Three main conclusions emerge:

(1) Few-shot prompting is unreliable for classification but conditionally useful for extraction. In classification, adding in-context examples hurt most models, with PaliGemma2-3B suffering a huge mode collapse. Only CLIP benefited, since its contrastive architecture genuinely uses the example labels rather than attending over image tokens. For extraction, the pattern is reversed: large models (Qwen2-VL-7B, Qwen2.5-VL-3B) improve their character F1 with 10-shot examples, while small models (SmolVLM-500M, PaliGemma2-3B) suffer from context saturation or formatting instability. The practical implication is that the optimal inference strategy is task- and model-dependent, and that blindly applying few-shot prompting may be counterproductive.

(2) Fine-tuning with limited data and parameter-efficient methods closes most of the gap to specialist models. PaliGemma2-3B with LoRA reaches 90.0% classification accuracy on RVL-CDIP, only 3 p.p. below the specialist DiT-large. DeiT-small, with just 22M parameters, beats all zero-shot and few-shot VLMs while being far cheaper to run and store. Both models were trained on 32,000 images (a realistic data budget for enterprise settings) in a resource-constrained GPU environment. This suggests that targeted fine-tuning on a company-specific document taxonomy is the most practical path toward production reliability.

(3) Qwen2-VL-7B is the best off-the-shelf model for both tasks. It leads zero-shot classification among VLMs (54.6%), achieves the highest invoice F1 in zero-shot (0.693), and is the strongest extractor in both zero-shot (CF1 = 0.820) and few-shot (CF1 = 0.888) settings. It is the only model that improves consistently across metrics when few-shot examples are added to extraction prompts. Moreover, its 7B parameter size strikes a favorable balance between performance and computational cost, making it a compelling candidate for future enterprise deployment in document intelligence pipelines.

6 Limitations and Future Work

While the experiments provide meaningful insights into the capabilities of open-source VLMs for document intelligence, a number of limitations remain. These concern both the project setup and the gap between the controlled research environment and a real-world enterprise deployment, and are discussed below alongside directions for future work.

Resource-constrained setting. All experiments ran under GPU memory constraints typical of a research environment rather than a production deployment. Larger-batch training, full-precision inference, and evaluation on the full RVL-CDIP test set could shift results. Furthermore, models close to our hardware limits (Qwen2-VL-7B) may have been quantised or partially offloaded in ways that can affect accuracy.

Data quality and distribution shift. RVL-CDIP contains scanned documents from a North American archive of the 1990s–2000s. SROIE contains Malaysian retail receipts. Neither dataset reflects the multilingual, regulation-compliant, and digitally-born PDFs that a specifically-designed platform would process. Performance may differ substantially on real client documents, particularly for address extraction and company name recognition where language and formatting conventions are highly country-specific.

Extraction task fine-tuning. We only fine-tuned models for the classification task. Applying LoRA to Qwen2-VL-7B for structured field extraction — using a dataset of annotated Swiss invoices — is a natural next step and could substantially improve company name and address accuracy, the two fields where VLMs already outperform the regex baseline but have the most room to grow.

Reliability and confidence scoring. For financial documents, the cost of extraction errors is asymmetric: a missed digit in a `total` field has direct monetary consequences. A safe production system should include a confidence score alongside each prediction, enabling automatic routing to human review when the model is uncertain. None of the models tested provide calibrated probabilities out of the box. On the whole, a human-in-the-loop architecture would compensate for model errors at critical stages of the workflow.

End-to-end pipeline integration. Classification and extraction were evaluated independently in this project. A realistic pipeline would chain them: classify to determine the document type, then deliver to a type-specific extraction model. Error propagation across this chain – and the design of fallback strategies when classification is uncertain – has not been studied and represents major future engineering work.

References

[1] A. W. Harley, A. Ufkes, and K. G. Derpanis, *Evaluation of Deep Convolutional Nets for Document Image Classification and Retrieval*, ICDAR, 2015.

[2] Z. Huang et al., *ICDAR2019 Competition on Scanned Receipt OCR and Information Extraction*, ICDAR, 2019.

[3] J. Li et al., *DiT: Self-supervised Pre-training for Document Image Transformer*, ACM MM, 2022.

[4] H. Touvron et al., *Training Data-Efficient Image Transformers & Distillation through Attention*, ICML, 2021.

[5] L. Beyer et al., *PaliGemma 2: A Family of Versatile VLMs for Transfer*, arXiv:2412.03555, 2024.

[6] P. Wang et al., *Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution*, arXiv:2409.12191, 2024.

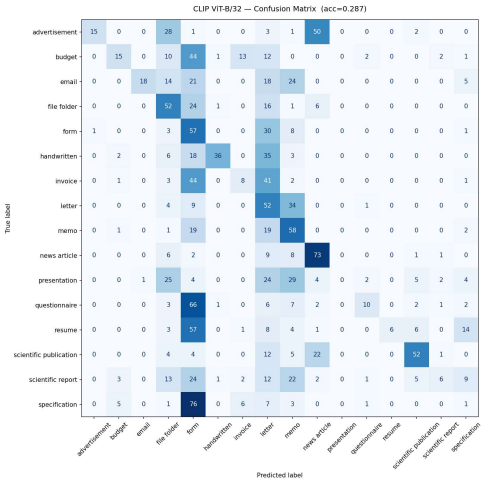
[7] A. Radford et al., *Learning Transferable Visual Models From Natural Language Supervision*, ICML, 2021.

[8] HuggingFace, *SmolVLM: Redefining Small and Efficient Multimodal Models*, `https://huggingface.co/blog/smolvlm2`, 2024.

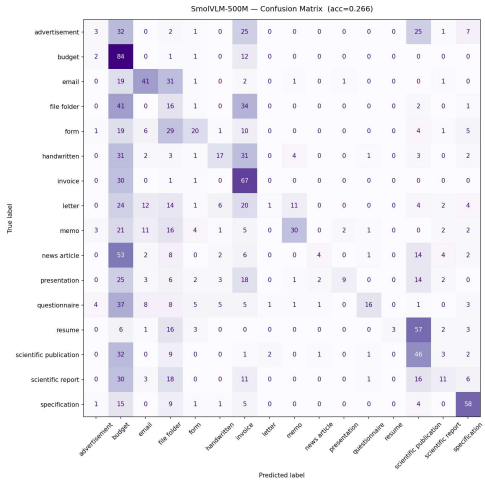
[9] E. J. Hu et al., *LoRA: Low-Rank Adaptation of Large Language Models*, ICLR, 2022.

A Additional Figures

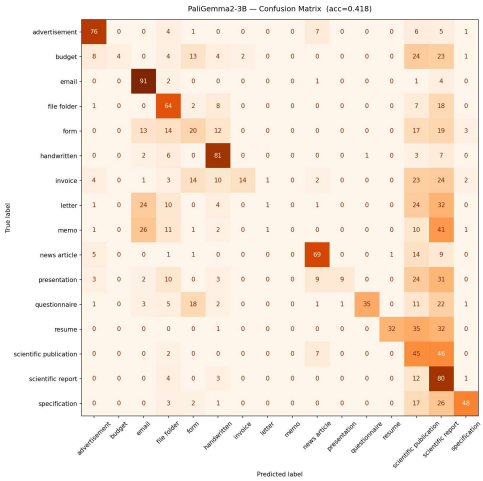
A.1 Zero shot



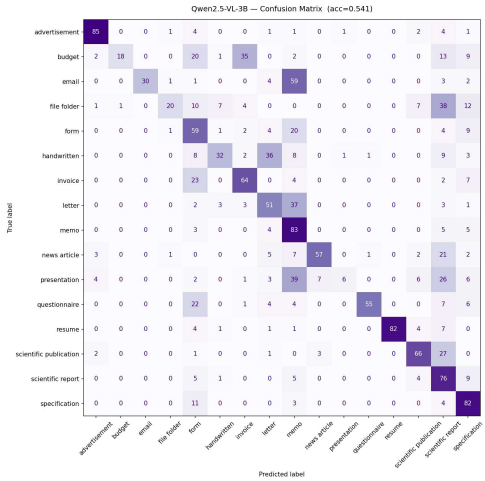
(a) CLIP



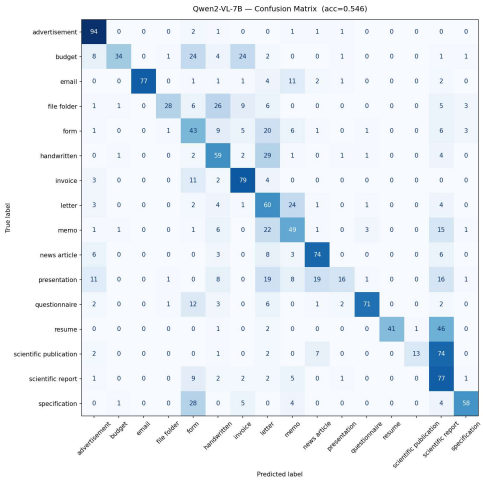
(b) SMOL-vLM



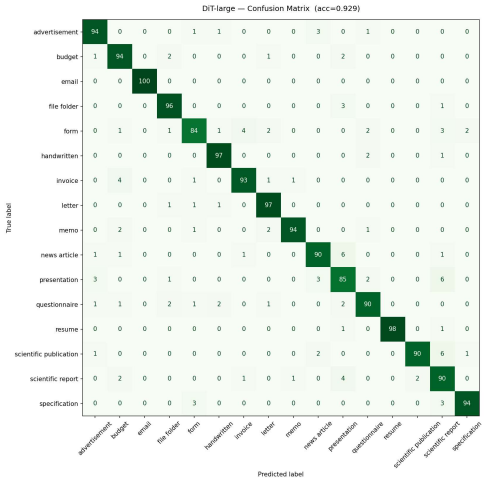
(c) Paligemma 3b



(d) Qwen 2.5 VL 3b

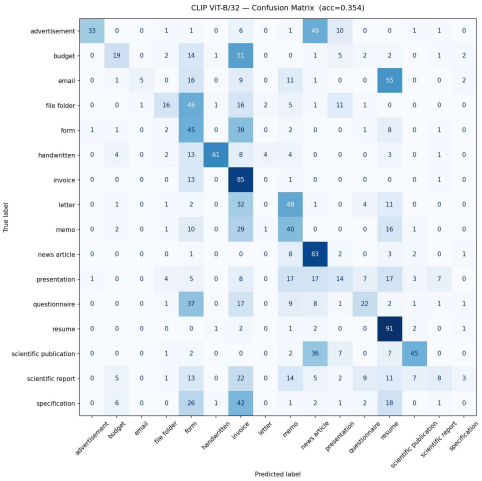


(e) Qwen 2 VL 7b

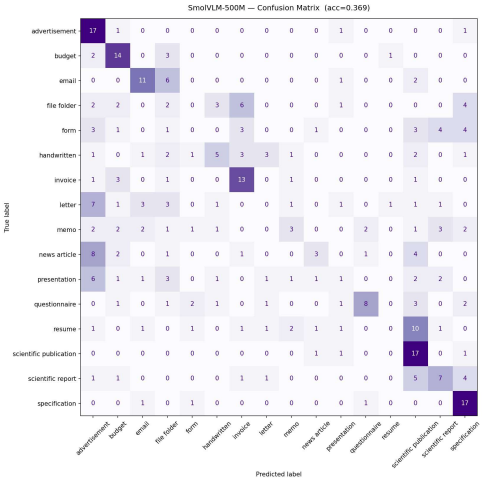


(f) DiT Large

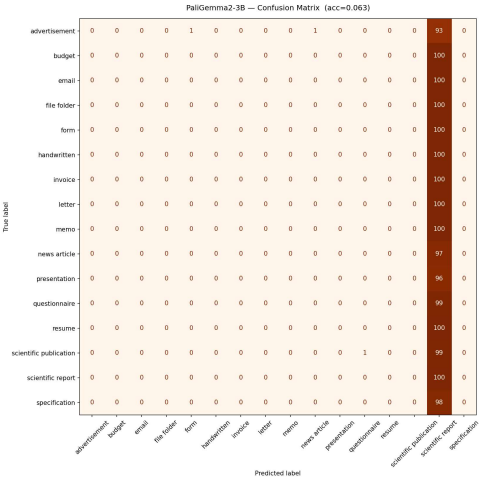
A.2 Few shot



(a) CLIP

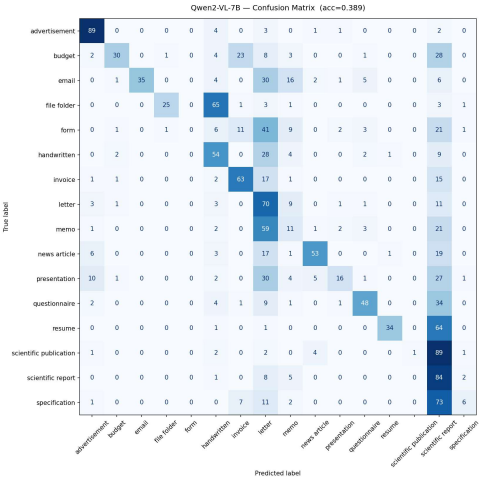


(b) SMOL-vLM

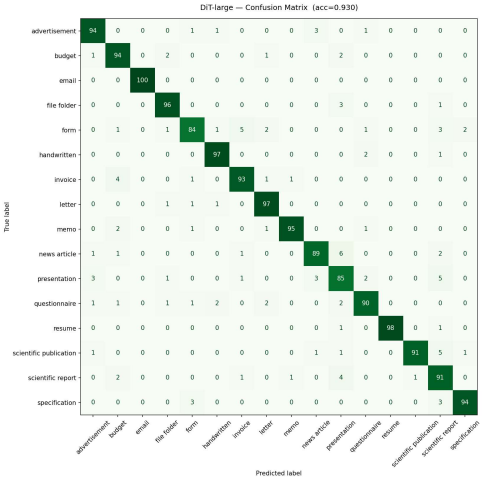


(c) Paligemma 3b

(d) Qwen 2.5 VL 3b



(e) Qwen 2 VL 7b



(f) DiT Large

A.3 Few-Shot vs. Zero-Shot Performance Delta

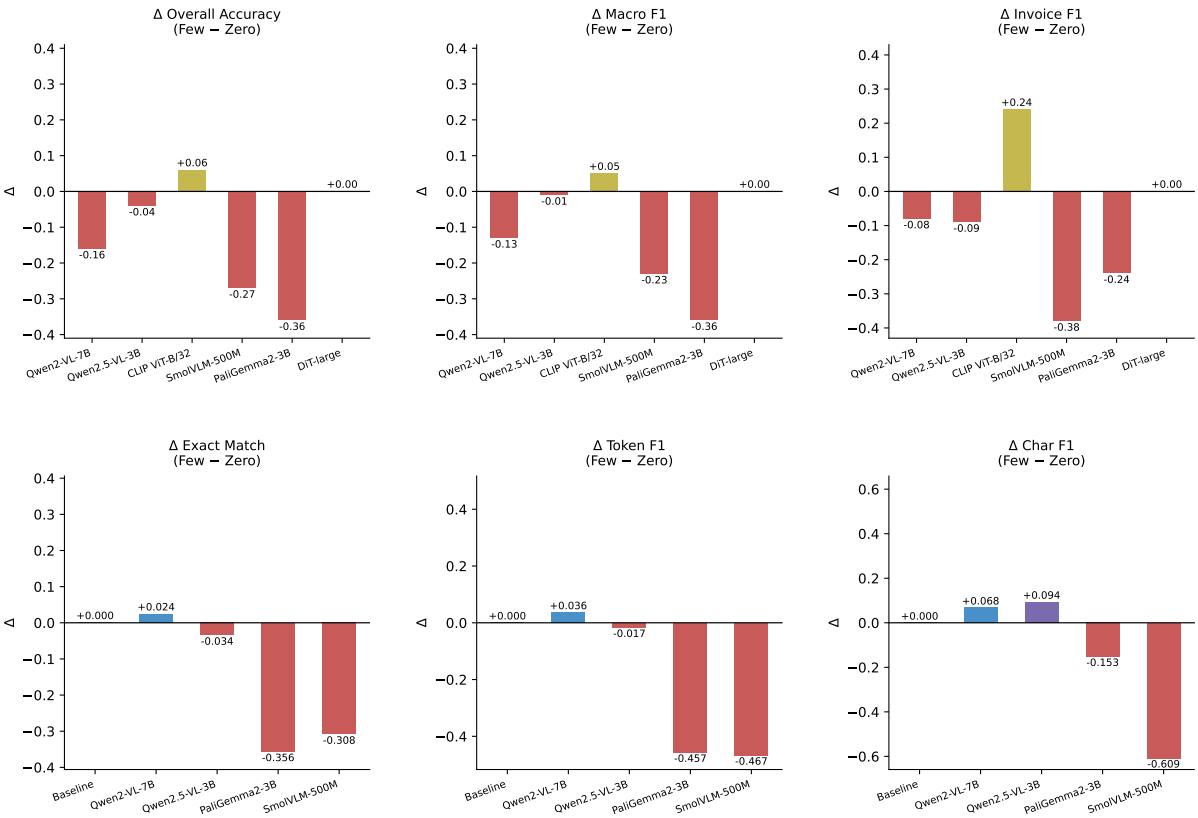
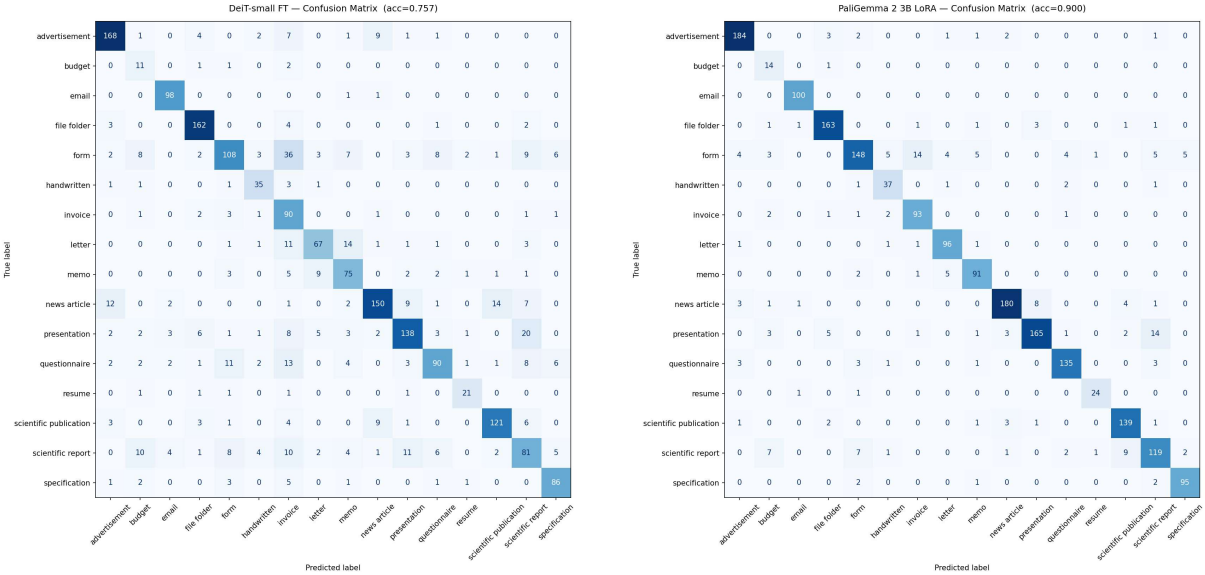


Figure 10:
Delta for classification (top row) and extraction (bottom row). Positive values indicate few-shot helped; negative values indicate it hurt.

A.4 Fine-Tuning: Confusion Matrices and Per-Class Analysis



(a) DeiT-small (fine-tuned). (b) PaliGemma2-3B LoRA (fine-tuned).

Figure 11: Fine-tuned confusion matrices. PaliGemma2-3B LoRA shows notably cleaner per-class separation.

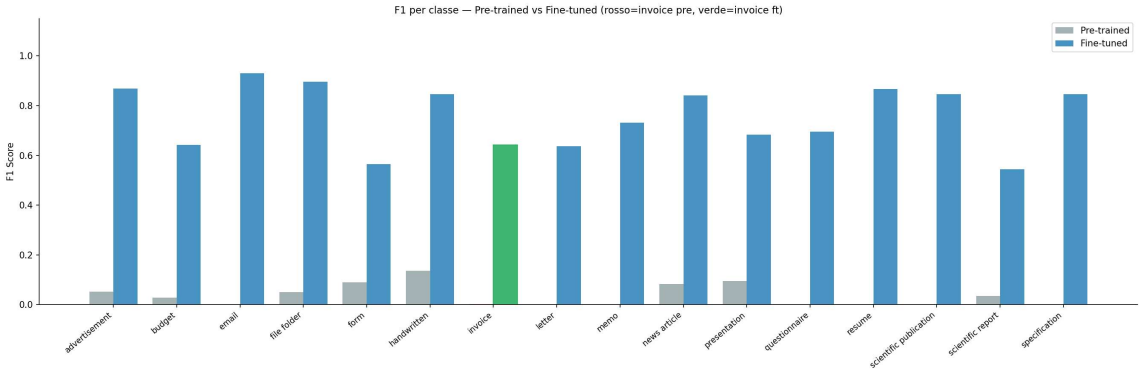


Figure 12: F1 per class: DeiT-small pre-trained vs. fine-tuned.

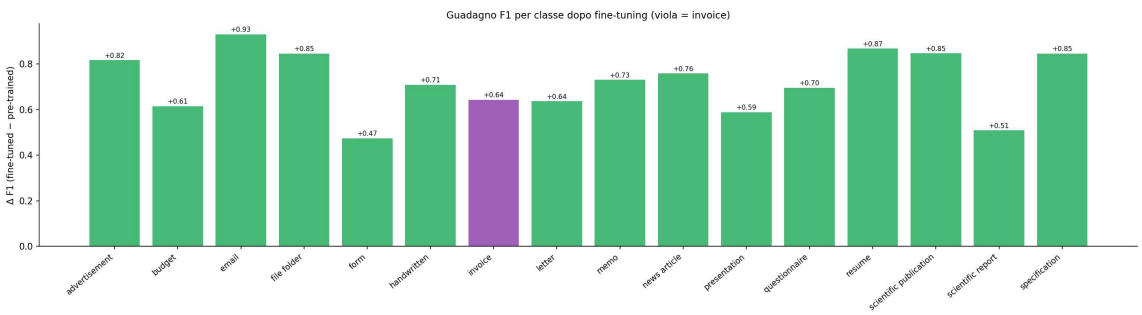


Figure 13: Delta F1 per class from fine-tuning.